# Benjamin V. Lehmann

**Einstein Fellow,** NASA Hubble Fellowship Program, 2026–2029. [tinyurl.com/bvlnhf]
**Pappalardo Fellow,** MIT Center for Theoretical Physics, 2022–2025. [tinyurl.com/bvlppf]

Department of Physics & Kavli Institute for Astrophysics and Space Research
Massachusetts Institute of Technology.


(Updated October 27, 2026)

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## Education

Ph. D., Physics. *Advisor:* Stefano Profumo. August 2022
University of California, Santa Cruz.

M. S., Mathematics. *Advisor:* Eleny Ionel. June 2016
Stanford University.

B. S., Mathematics; B. S., Physics, with honors. *Advisor:* Risa Wechsler. June 2015
Stanford University.

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## Selected awards

Michelson Postdoctoral Prize Lectureship. 2026
*A competitive prize awarded to one postdoctoral scholar across all fields of physics.*

Infinite Expansion Award, Massachusetts Institute of Technology. 2025
*An award presented by the School of Science to "extraordinary members of the MIT community."*

Josephine de Karman Dissertation Fellowship. 2021–2022
*A highly competitive dissertation-year fellowship open to applicants from all fields.*

Honorable Mention, Ford Foundation Dissertation Fellowship competition. March 2021
Ford Foundation & National Academies of Science, Engineering, and Medicine.

Steck Family Prize for Graduate Research Excellence, UC Santa Cruz. April 2019

ARCS Scholarship. 2018–2019
*Only the department's best student is eligible for nomination for this significant award.*

Marilyn Stevens Memorial Award, UC Santa Cruz Department of Physics. June 2018
*This award is granted for outstanding departmental service in addition to academic excellence.*

Fridley Scholarship Award, UC Santa Cruz. May 2017
*The Fridley Scholarship is awarded to the single best student in the physical sciences at UCSC.*

Regents' Fellowship, UC Santa Cruz. September 2016–September 2017

# Seminars

Cosmology Seminar, Tufts University. November 2026
*X-ray femtolensing probes asteroid-mass black holes.*

Seminar, University of Tennessee, Knoxville. October 2026
*Broadening direct searches for light dark matter.*

Seminar, ICTP-AP. September 2026
*Broadening direct searches for light dark matter.*

Seminar, Brookhaven National Laboratory. July 2026
*Broadening direct searches for light dark matter.*

**Colloquium**, Case Western Reserve University. April 2026
*Testing the Particle Dark Matter Paradigm.*

Seminar, Case Western Reserve University. April 2026
*Compact objects as dark matter: Endgame.*

Seminar, Case Western Reserve University. April 2026
*Broadening direct searches for light dark matter.*

Seminar, Harvard University. February 2026
*Broadening direct searches for light dark matter.*

Seminar, Technion – Israel Institute of Technology, Israel. December 2025
*New tools for dark matter physics.*

Seminar, University of Michigan. November 2025
*Broadening direct searches for light dark matter.*

Seminar, IFAE / Universitat Autònoma de Barcelona. October 2025
*Broadening direct searches for light dark matter.*

KCTP Seminar, University of Chicago. May 2025
*Broadening direct searches for light dark matter.*

Seminar, Rutgers University New High Energy Theory Center. March 2025
*New tools for dark matter physics.*

Theory Seminar, SLAC. February 2025
*New tools for dark matter physics.*

Seminar, Boston University. February 2025
*Finding primordial black holes wherever they may hide.*

Seminar, Rutgers University New High Energy Theory Center. February 2025
*Finding primordial black holes wherever they may hide.*

Seminar, Kavli Institute for Particle Astrophysics and Cosmology. November 2024
*Finding primordial black holes wherever they may hide.*

IDEA Seminar, Brown University. November 2024
*Finding primordial black holes wherever they may hide.*

Theory Seminar, Washington University. November 2024
*Finding primordial black holes wherever they may hide.*

Seminar, Oklahoma State University. October 2024
*Broadening direct searches for light dark matter.*

SEMINAR, University of Oklahoma. October 2024
*Broadening direct searches for light dark matter.*

SOTU SEMINAR, Tata Institute of Fundamental Research. October 2024
*Broadening direct searches for light dark matter.*

SEMINAR, University of California, Santa Cruz. April 2024
*Direct detection of dark matter far from the weak scale.*

SEMINAR, Texas A&M University. March 2024
*Direct detection of dark matter far from the weak scale.*

SEMINAR, Queen's University. February 2024
*Direct detection of dark matter far from the weak scale.*

SEMINAR, Dark Matter: First Light Workshop, Perimeter Institute for Theoretical Physics. February 2024
*Kinetic recoupling of dark matter.*

SEMINAR, Kavli Institute for the Physics and Mathematics of the Universe. January 2024
*Direct detection of dark matter far from the weak scale.*

SEMINAR, DMPNP Workshop, Asia-Pacific Center for Theoretical Physics. January 2024
*New directions for direct detection with dielectrics.*

SEMINAR, University of Florida. November 2023
*New directions for direct detection with dielectrics.*

SEMINAR, University of Illinois Urbana-Champaign. March 2023
*Kinetic recoupling of dark matter.*

KICP SEMINAR, University of Chicago. March 2023
*New frontiers in dark matter direct detection.*

SEMINAR, Fermilab Cosmic Physics Center. March 2023
*Kinetic recoupling of dark matter.*

HETG FAMILY MEETING SEMINAR, Harvard University. March 2023
*Developments in Directional DM detection in dielectrics.*

PROGRAM SEMINAR, Simons Center for Geometry and Physics. February 2023
*Directional DM detection in dielectrics.* [tinyurl.com/bddjxdbr]

SEMINAR, University of Texas at Austin. February 2023
*Kinetic recoupling of dark matter.*

HETG SEMINAR, Harvard University. January 2023
*Kinetic recoupling of dark matter.*

MIT/TUFTS COSMOLOGY SEMINAR December 2022
*Kinetic recoupling of dark matter.*

SEMINAR, Cornell University. November 2022
*Direct detection of dark matter far from the weak scale.*

SEMINAR, McGill University, Canada. October 2022
*Direct detection of dark matter far from the weak scale.*

SEMINAR, California Institute of Technology. October 2022
*Direct detection of dark matter far from the weak scale.*

SEMINAR, K. N. Toosi University of Technology, Iran,*. July 2022
*Primordial black holes and extrasolar systems.*

Seminar, Weizmann Institute of Science, Israel. May 2022
*New physics from black holes with present-day tools.*

Seminar, Technion – Israel Institute of Technology, Israel. May 2022
*Direct detection of dark matter far from the weak scale.*

Particle Physics Joint Seminar, Hebrew University, Israel. April 2022
*Direct detection of dark matter far from the weak scale.*

EPT Seminar, University of Maryland. February 2022
*Direct detection of dark matter far from the weak scale.*

**Theory Colloquium**, CU Boulder. February 2022
*Direct detection of dark matter far from the weak scale.*

Seminar, BSM PANDEMIC Series. December 2021
*Massive black hole binaries as particle physics laboratories.*

KIPAC Tea Talk, Stanford University. November 2021
*Massive black holes as particle physics laboratories.*

HEP Seminar, UC San Diego. September 2021
*Extracting new physics from black holes with present-day tools.*

4D Seminar, Berkeley Center for Theoretical Physics. September 2021
*Direct detection of dark matter far from the weak scale.*

Seminar, NANOGrav Astrophysics Working Group Meeting. June 2021
*Detecting new forces in the gravitational wave background.*

Seminar, SuperCDMS Science Meeting. April 2021
*Dark matter–electron scattering and the dielectric function.*

Seminar, Fermilab Cosmic Physics Center. March 2021
*Discovering new physics in the era of black holes.* [tinyurl.com/e36mb94w]

HEP Seminar, University of Notre Dame. October 2020
*New opportunities for primordial black holes across the scales.*

Joint UC Santa Cruz–UC Irvine theory seminar May 2020
*FIMP dark matter at the KOTO experiment.*

SCIPP Seminar, Santa Cruz Institute for Particle Physics. January 2020
*Catching black holes in the lab.*

Cosmology Seminar, UC Santa Cruz. June 2016
*Some Assembly Required: Modeling Assembly Bias with Abundance Matching.*

# Selected conference presentations

Contributed talk, Hubble Fellows Symposium. October 2026
*X-ray femtolensing probes asteroid-mass black holes.*

Contributed talk, MITP Program: The Phenomenology of Primordial Black Holes. July 2026
*X-ray femtolensing probes asteroid-mass black holes.*

Contributed talk, Black Holes & Cosmology Workshop. June 2026
*X-ray femtolensing probes asteroid-mass black holes.*

**Invited overview**, particle dark matter mini-symposium, APS Global Physics Summit. March 2026
*The Present and Future of Particle Dark Matter.*

**Invited talk**, Rainer Weiss Memorial Symposium. February 2026
*The final frontier for black holes as dark matter.*

**Invited plenary**, SEARCH25, CERN. October 2025
*Theory opportunities—is there any room for WIMPs?*

Workshop talk, Aspen Center for Physics. August 2025
*Physics from gravitational wave propagation.*

Contributed talk, Dark Side of the Universe. July 2025
*New dynamical probes of primordial black holes.*

Contributed talk, Aspen Winter Conference. March 2025
*Direct detection: Materials physics for discovery.*

**Invited talk**, Miami Conference. December 2024
*Broadening direct searches for light dark matter.*

Contributed talk, TeVPA. August 2024
*Broadening direct searches for light dark matter.*

Contributed talk, APS DPF / Phenomenology Symposium. May 2024
*Close encounters of the primordial kind:*
*a new observable for primordial black holes as dark matter.*

Contributed talk, Black Holes and Cosmology Workshop. March 2024
*Detecting new forces in the gravitational wave background.*

Contributed talk, PLANCK. May 2023
*Directional detection of dark matter with anisotropic response functions.*

Contributed talk, COSMO'22. August 2022
*Direct detection with superconducting nanowires.*

Contributed talk, Dark Sectors of Astroparticle Physics (AstroDark-2021). December 2021
*Detecting new forces in the gravitational wave background.*

Contributed talk, APS Division of Particles and Fields Meeting (DPF). July 2021
*Detecting new forces in the gravitational wave background.*

Contributed talk, Cosmology From Home. July 2021
*Detecting new forces in the gravitational wave background.* [youtu.be/iSQD4i-sjMA]

Contributed talk, Phenomenology 2021 Symposium (Pheno). May 2021
*Detecting new forces in the gravitational wave background.*

Contributed talk, PPC. May 2021
*Detecting new forces in the gravitational wave background.*

Poster, "A Rainbow of Dark Sectors" Conference (Aspen Center for Physics). March 2021
*Detecting new forces in the stochastic gravitational wave background.*

Contributed talk, International Conference on New Frontiers in Physics (ICNFP). October 2020
*FIMP dark matter at the KOTO experiment.*

Contributed talk, TASI student talks. June 2020
*Direct detection of primordial black hole dark matter.*

Contributed talk, Phenomenology Symposium (Pheno). May 2020
*FIMP dark matter at the KOTO experiment.*

Contributed talk, SUSY. May 2019
*Dark sources of cosmic ray spikes on Earth's doorstep.*

Contributed talk, Phenomenology Symposium (Pheno). May 2019
*Direct detection of primordial black hole dark matter.*

Contributed talk, SUSY. July 2018
*Sub-MeV Dark Matter Interactions with Electrons: Cosmological Constraints and Detection Prospects*

Contributed talk, Phenomenology Symposium (Pheno). May 2018
*Effective field theory and cosmology of sub-MeV leptophilic dark matter.*

Poster, SLAC Summer Institutes (SSI), SLAC. August 2017
*Cosmological constraints and detection prospects for sub-MeV dark matter.*

# Selected outreach

Volunteer participant, Skype a Scientist [skypeascientist.com]. October 2019–present
*This program connects scientists with classrooms for live conversations. I have spoken with 13 groups.*

Volunteer participant, Letters to a Pre-Scientist [prescientist.org]. October 2022–present
*This pen-pal program connects STEM professionals with middle school students from all backgrounds. I have exchanged letters with 4 students.*

Volunteer instructor, The Educational Justice Institute at MIT [teji.mit.edu]. October 2022–2024
*Volunteer instructor in the Massachusetts correctional system.*

Physics Collaborative facilitator, UC Santa Cruz Department of Physics. January 2019–June 2022
*Mentor for students in introductory physics courses.*

SPS/WiPA program mentor, UC Santa Cruz Department of Physics. January 2018–June 2022
*Program matches undergraduates with graduate students for long-term mentoring.*

Volunteer instructor, UCSC Project for Inmate Education (PIE). October 2016–June 2022
*Volunteer instructor for mathematics and astronomy courses at Santa Cruz County Jail.*

Peer mentor, SU(5) [su5.group]. August 2020–2021
*SU(5) was a peer mentorship organization supporting incoming physics graduate students.*

Volunteer instructor. August 2015–December 2016
Prison University Project / Mount Tamalpais College [mttamcollege.edu].
*Volunteer instructor for mathematics courses in a degree program at San Quentin State Prison.*

### Media

Scientific expert for *Science News Explores* magazine [snexplores.org]. 2025–present

Scientific advisor for PBS NOVA series *Decoding the Universe*. 2024

# Students and mentoring

**Current students**

I am currently supervising part or all of the research work of the following students:

Joheen Chakraborty (Ph. D., MIT);
Guy Daniel Hadas (Ph. D., Hebrew U.);
Omer Tzvi Katz (Ph. D., Tel Aviv U.).
Jackie Lodman (Ph. D., Harvard U.);
Rotem Ovadia (Ph. D., Hebrew U.);

**Undergraduate theses supervised**

I directly supervised the following undergraduate theses under the oversight of my Ph. D. advisor.

Samuel D. English, January 2020–June 2022.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), University of New Mexico.
*The Hidden Friend's Wake: Dark Matter and a Binary at the Galactic Center.*
Selected for **Honors** in the major.

Olive Ross, November 2018–June 2021.
B. S., UC Santa Cruz ⟶ Ph. D. (Astronomy), Cornell University.
*Searching for primordial black holes.*
Selected for **Honors** in the major.
**Steck Family Award** for the single best senior thesis of 2020–2021 at UC Santa Cruz.

Paul Andreini, March 2017–September 2019.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), University of Oregon.
*Time-evolution of the primordial black hole mass function.*
Selected for **Honors** in the major.

Thomas Schwemberger, February 2018–December 2018.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), University of Oregon.
*Constraints on Planck mass relics from primordial black hole evaporation with a generic initial mass function.*
Selected for **Honors** in the major.

Anikeya Aditya, September 2016–June 2018.
B. S., UC Santa Cruz ⟶ Ph. D. (Materials Science), University of Southern California.
*Inferring the mass function of primordial black holes from gravitational wave observations.*
Selected for **Honors** in the major.

**Other former student collaborators**

Badal Bhalla, October 2022–May 2026.
Ph. D., University of Oklahoma ⟶ Postdoctoral Associate, University of Utah.

Bethany Suter, May 2023–May 2026.
Ph. D., UC Berkeley ⟶ Postdoctoral Associate, Fermilab.

Marcell Howard, January 2024–May 2026.
Ph. D., University of Pittsburgh ⟶ Postdoctoral Associate, Brown University.

Christina Schwemmbauer, April 2022–October 2025.
Ph. D., DESY ⟶ Photonic Team Lead, Munich Quantum Instruments.

Cara Giovanetti, January 2024–May 2025.
Ph. D., NYU ⟶ Postdoctoral Fellow, UC Berkeley.

Wenzer Qin, November 2023–May 2024.
Ph. D., MIT ⟶ Simons Junior Fellow, NYU.

Sarah R. Geller, January 2023–May 2024.
Ph. D., MIT ⟶ NSF MPS Postdoctoral Fellow, UC Santa Cruz / UC Berkeley.

Tung X. Tran, January 2023–May 2024.
B. S., MIT ⟶ Ph. D. (Physics), Stanford University.

Nolan Smyth, January 2021–June 2024.
Ph. D., UC Santa Cruz ⟶ Postdoctoral Fellow, Mila / Université de Montréal.

Ava Webber, September 2020–June 2022.
B. S., UC Santa Cruz ⟶ Ph. D. (Astronomy), Clemson University.

Mason Hargrave, September 2018–June 2019.
B. S., UC Santa Cruz ⟶ Ph. D. (Computational Biology), Rockefeller University.

Jackson Yant, January 2017–June 2018.
B. S., UC Santa Cruz ⟶ Ph. D. (Physics), Dartmouth College.

# Professional activities

**Referee** for Physical Review Letters (**PRL**), Physical Review D (**PRD**), Journal of High Energy Physics (**JHEP**), Journal of Cosmology and Astroparticle Physics (**JCAP**), Physics Letters B (**PLB**), European Physical Journal C (**EPJC**), The Astrophysical Journal (**ApJ**), Astrophysics and Space Science (**Ap&SS**), and Physics of the Dark Universe (**PDU**).

**Scientific Organizing Committee member,** Hubble Fellows Symposium. 2026

**Subject Expert** for Berggruen Prize Essay Competition. 2025

### Other workshops and schools

Aspen Summer Workshop: *Physics in the Era of Nanohertz Gravitational Waves.* August 2025
Aspen Center for Physics.

Aspen Winter Conference: *Empowering the New Vision in High Energy Physics.* March 2025
Aspen Center for Physics.

KITP Program: June–July 2024
*Dark Matter Theory, Simulation, and Analysis in the Era of Large Surveys.*
Kavli Institute for Theoretical Physics.

APCTP Workshop: *Dark Matter as a Portal to New Physics (DMPNP).* January 2024
Asia-Pacific Center for Theoretical Physics.

GUINEAPIG Workshop, University of Montreal. July 2023

MITP Workshop: *Theoretical Particle Cosmology in the Early and Late Universe.* May 2023
MITP & Corfu Summer Institute.

SCGP Program: March 2023
*Lighting new Lampposts for Dark Matter and Beyond the Standard Model.*
Simons Center for Geometry and Physics.

EXCESS2022 Workshop, Karlsruhe Institute of Technology. February 2022

Les Houches Summer School 2021: *Dark Matter.* July–August 2021
École de Physique des Houches.

TASI 2020: *The Obscure Universe: Neutrinos and other Dark Matters.* June 2020
Theoretical Advanced Study Institute.

37th Advanced School in Theoretical Physics: December 2019–January 2020
*New Ideas for Old Puzzles in Particle Physics.*
Israel Institute for Advanced Studies.

GGI Workshop: *Next Frontiers in the Search for Dark Matter.* August–September 2019
Galileo Galilei Institute.

45th SLAC Summer Institute: *Cosmic Opportunities.* July 2017
SLAC National Accelerator Laboratory.

**Committee service**

Physics Values Committee. **Committee chair,** November 2023–present. October 2022–present
Department of Physics, Massachusetts Institute of Technology.
[physvals.mit.edu]

Nuclear and Particle Theory Seminar Committee. August 2024–May 2025
Department of Physics, Massachusetts Institute of Technology.

Diversity and Climate Committee. February 2018–June 2022
Department of Physics, UC Santa Cruz.
[www.physics.ucsc.edu/about/diversity/diversity-committee.html]

Colloquium Committee. June 2018–June 2022
Department of Physics, UC Santa Cruz.

CUWiP 2022/2023 Local Organizing Committee at UC Santa Cruz. January 2020–January 2023
Conference for Undergraduate Women in Physics.

# Teaching experience

Selected student comments are provided for all courses in which I received any narrative feedback. Due to platform changes, student evaluations are not available between Spring 2019 and Spring 2020.

**Teaching assistant, UC Santa Cruz.**

Introductory Physics II (PHYS 5B). March–June 2021
*"It's difficult to explain how good of a TA Ben actually was. [...] His teaching method was spot on."*

Thermodynamics and Statistical Mechanics (PHYS 112). January–March 2021
*"Ben is a very kind and inclusive person, great at mentoring and teaching..."*
*"I really appreciated the feeling of support that Ben gave us as students..."*
*"He's the goat and deserves a raise forreal."*

General Chemistry III (CHEM 1C). September–December 2020
*"Best TA I ever had."*

Particle Astrophysics (PHYS 129). January–April 2020

General Chemistry III (CHEM 1C). September–December 2019

Introductory Physics I Laboratory (PHYS 6L). April–June 2019

Computational Physics (PHYS 115). April–June 2019

Conceptual Physics (PHYS 1). January–April 2019
*"Everything I learned about physics this quarter I owe to Wednesday sections with Ben."*

Mathematical Methods in Physics III (PHYS 116C). September–December 2018
*"The best TA ever!"*

Quantum Mechanics I (PHYS 139A). April–June 2018
*"He knew when to give me answers versus [...] to suggest that I struggle a little more."*
*"Ben's awesome."*

Physics and Psychophysics of Music (PHYS / MUSC 80U). January–March 2018
*"Ben lets students think for themselves."*

**Lead TA**, Introductory Physics I (PHYS 6A). September–December 2017
*"Engaging, helpful, and extraordinarily friendly."*
*"Ben is good at teaching in a way that is easy to understand."*
*"Ben was clear and taught you how to think in physics."*

Introductory Physics I Laboratory (PHYS 6L). June–August 2017
*"Really enjoyed the way Ben taught the lab, he was awesome!"*

Introductory Physics I Laboratory (PHYS 6L). April–June 2017
*"I know I can never be as patient as him even next life."*
*"Everyone in class loves Ben."*

Introductory Physics II Laboratory (PHYS 6M). January–March 2017
*"Ben is one of the best TAs I have ever had."*

Introductory Physics I Laboratory (PHYS 6L). September–December 2016
*"When we were completely confused, he helped point us in the right direction."*

**Mount Tamalpais College / Prison University Project.**

**Lead instructor**, Intermediate Algebra (MATH 115). May–August 2016

Co-instructor, Elementary Algebra (MATH 99). January–April 2016

Study group leader, Elementary Algebra (MATH 99). August–December 2015

**Teaching assistant, Stanford University.**

Electricity, Magnetism, and Optics (PHYSICS 23). January–March 2014
*"Because of [Ben] I am actually planning on getting a minor in physics."*
*"I never asked a question that Ben couldn't give a good answer to."*
*"Good section leader who obviously cared a lot about us."*

# Publications

An up-to-date list of publications can be found at http://inspirehep.net/authors/1737895.
The majority of my publications follow the HEP convention, in which author order is alphabetical.

★ I was the project leader
■ I was the senior author
† I was an equal contributor (alphabetical author order)

[1] ■ **BVL**, Jackie Lodman, and Thomas Steingasser. Tunneling and tidal stripping in multifield ultralight dark matter halos. *JHEP*, May 2026, 2605.12589. In press.

[2] ★ Sinéad M. Griffin, Guy Daniel Hadas, Yonit Hochberg, Katherine Inzani, and **BVL**. Dark Matter–Electron Detectors for Dark Matter–Nucleon Interactions. *Phys. Rev. Lett.*, 135(14):141803, September 2025, 2412.16283. DOI 10.1103/6qqv-rl7q.

[3] ★ Badal Bhalla, **BVL**, Kuver Sinha, and Tao Xu. Fast and Fewrious: Stochastic binary perturbations from fast compact objects. *Phys. Rev. D*, 113(4):043040, July 2025, 2507.16894. DOI 10.1103/k56g-tf9j.

[4] ★ Sinéad M. Griffin, Yonit Hochberg, **BVL**, Rotem Ovadia, Kristin A. Persson, Bethany A. Suter, Ruo Xi Yang, and Wayne Zhao. First High-Throughput Evaluation of Dark Matter Detector Materials. *Phys. Rev. Lett.*, 136(19):191801, June 2025, 2506.19905. DOI 10.1103/bb4g-7939.

[5] † Christina Schwemmbauer, **BVL**, et al. First direct search for light dark matter interactions in a transition-edge sensor. *Phys. Rev. D*, 114(1):012006, June 2025, 2506.18982. DOI 10.1103/zytz-n25v.

[6] ★ **BVL**. Higher multipoles of the cow. *Am. J. Phys.*, April 2025, 2504.00506. A tongue-in-cheek but rigorous extension of the spherical cow approximation. **Featured Article**. [**AIP Scilight**]. DOI 10.1119/5.0287411.

[7] † Laura Baudis, **BVL**, et al. A New Bite Into Dark Matter with the SNSPD-Based QROCODILE Experiment. *Phys. Rev. Lett.*, 135:081002, December 2024, 2412.16279. [**Featured in Physics**] [**ScienceDaily**]. DOI 10.1103/4hb6-f6jl.

[8] ★ Badal Bhalla, **BVL**, Kuver Sinha, and Tao Xu. Dancing with invisible partners: Three-body exchanges with primordial black holes. *Phys. Rev. D*, 111(4):043029, August 2024, 2408.04697. [**New Scientist**] [**Astrobites**] DOI 10.1103/PhysRevD.111.043029.

[9] ★ Jiansong Gao, Yonit Hochberg, **BVL**, Sae Woo Nam, Paul Szypryt, Michael R. Vissers, and Tao Xu. Detecting Light Dark Matter with Kinetic Inductance Detectors. March 2024, 2403.19739. Submitted to *Physical Review Letters*.

[10] ★ Tung X. Tran, Sarah R. Geller, **BVL**, and David I. Kaiser. Close encounters of the primordial kind: a new observable for primordial black holes as dark matter. *Phys. Rev. D*, 110(6):063533, December 2023, 2312.17217. **Editors' Suggestion**. [**Featured in Physics**] [**LA Times**] [**Astrobites**] [**SciAm**] DOI 10.1103/PhysRevD.110.063533.

[11] ★ **BVL**, Logan Morrison, Stefano Profumo, and Nolan Smyth. Kinetic recoupling of dark matter. *JCAP*, 07:049, October 2023, 2310.20513. DOI 10.1088/1475-7516/2024/07/049.

[12] ★ Yonit Hochberg, Eric David Kramer, Noah Kurinsky, and **BVL**. Directional detection of light dark matter in superconductors. *Phys. Rev. D*, 107(7):076015, September 2023, 2109.04473. DOI 10.1103/PhysRevD.107.076015.

[13] † Wolfgang Altmannshofer, Stefania Gori, **BVL**, and Jianhong Zuo. UV physics from IR features: New prospects from top flavor violation. *Phys. Rev. D*, 107(9):095025, March 2023, 2303.00781. DOI 10.1103/PhysRevD.107.095025.

[14] † Christian Boyd, Yonit Hochberg, Yonatan Kahn, Eric David Kramer, Noah Kurinsky, **BVL**, and To Chin Yu. Directional detection of dark matter with anisotropic response functions. *Phys. Rev. D*, 108(1):015015, December 2022, 2212.04505. DOI 10.1103/PhysRevD.108.015015.

[15] ★ Yonit Hochberg, **BVL**, Ilya Charaev, Jeff Chiles, Marco Colangelo, Sae Woo Nam, and Karl K. Berggren. New constraints on dark matter from superconducting nanowires. *Phys. Rev. D*, 106(11):112005, December 2022, 2110.01586. **Editors' Suggestion**. [**Featured in Physics**] DOI 10.1103/PhysRevD.106.112005.

[16] ★ **BVL** and Stefano Profumo. Black hole remnants are not too fast to be dark matter. *Phys. Dark Univ.*, 39:101145, November 2022, 2105.01627. DOI 10.1016/j.dark.2022.101145.

[17] ★ **BVL**, Ava Webber, Olivia G. Ross, and Stefano Profumo. Capture of primordial black holes in extrasolar systems. *JCAP*, 08:079, August 2022, 2205.09756. DOI 10.1088/1475-7516/2022/08/079.

[18] † Daniel Davies, Michael Dine, and **BVL**. Light Quarks at Large $N$. January 2022, 2201.05719. Contribution to a Festschrift for Peter Suranyi.

[19] † Jeff A. Dror, **BVL**, Hiren H. Patel, and Stefano Profumo. Discovering new forces with gravitational waves from supermassive black holes. *Physical Review D*, 104(8):083021, October 2021, 2105.04559. DOI 10.1103/PhysRevD.104.083021.

[20] † Yonit Hochberg, Yonatan Kahn, Noah Kurinsky, **BVL**, To Chin Yu, and Karl K. Berggren. Determining Dark Matter–Electron Scattering Rates from the Dielectric Function. *Phys. Rev. Lett.*, 127:151802, October 2021, 2101.08263. DOI 10.1103/PhysRevLett.127.151802.

[21] ★ **BVL**, Olivia G. Ross, Ava Webber, and Stefano Profumo. Three-body capture, ejection, and the demographics of bound objects in binary systems. *Monthly Notices of the Royal Astronomical Society*, 505(1):1017–1028, April 2021, 2012.05875. DOI 10.1093/mnras/stab1121.

[22] ★ **BVL**, Stefano Profumo, and Jackson Yant. Model-independent discovery prospects for primordial black holes at LIGO. *Monthly Notices of the Royal Astronomical Society*, 501(3):3727–3740, December 2020, 2007.00021. DOI 10.1093/mnras/staa3806.

[23] † Wolfgang Altmannshofer, **BVL**, and Stefano Profumo. Cosmological implications of the KOTO excess. *Physical Review D*, 102(8):083527, October 2020, 2006.05064. DOI 10.1103/PhysRevD.102.083527.

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